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Alzheimer's Disease Dataset Report Final

Alzheimer's disease is a progressive neurodegenerative disorder that affects memory, thinking, and behavior. It is the most common cause of dementia, with millions of individuals affected and numbers expected to rise significantly as populations age. Despite its prevalence, the causes and risk factors of Alzheimer's remain misunderstood. The lack of clarity makes early diagnosis, effective treatment, and targeted prevention challenging. As a result, Alzheimer's poses not only a medical challenge but also a socioeconomic challenge, especially for underserved communities with limited access to healthcare education of the disease.

The analysis of Alzheimer's disease data is essential to deepen our understanding of the factors that contribute to the onset and progression of the condition. Data-driven approaches can uncover hidden trends and correlations that may not be evident through clinical observation alone. By studying the relationship between age and cognitive decline, researchers can assess how aging influences brain function and work toward identifying

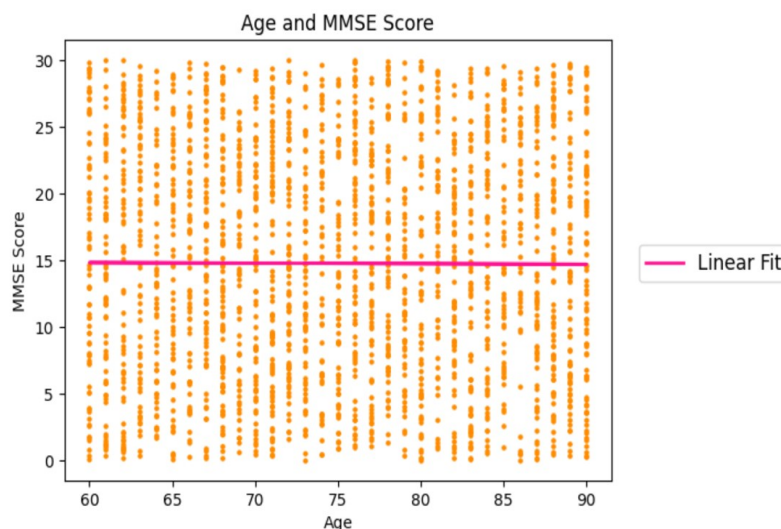
early biomarkers of cognitive impairment. Equally important is the examination of lifestyle factors such as smoking, exercise, alcohol consumption, diet, and sleep. These are modifiable behaviors that, if proven to influence disease risk, could empower individuals to take preventative actions. Understanding these associations also helps health professionals develop public health strategies aimed at lowering population-level risk. Additionally, analyzing education levels among patients contributes to the exploration of the cognitive reserve hypothesis (Rabie El Kharoua).

Together, these analyses provide valuable insights that can guide early intervention strategies, improve diagnostic accuracy, and inform policymakers. They also highlight the importance of addressing health inequalities, ensuring that vulnerable populations receive the attention and resources needed to mitigate the impact of Alzheimer's disease ("Data, Assessment, and Analysis").

To determine the correlation between age and MMSE score, the interpretation of an MMSE score needs to be introduced. MMSE, or Mini-Mental State Examination, can range in scores from 0 to 30, with scores under 25 indicating cognitive impairment (Health Direct). Using the Pandas, Scipy, and Matplotlib libraries, a scatterplot was created with age on the x axis and MMSE score on the y axis. A linear regression line was added and the correlation of the two variables was calculated. To figure out how smoking, drinking, physical activity, diet, and sleep effects Alzheimer's Disease diagnoses, the K Nearest Neighbor algorithm and a confusion matrix were utilized. The x data included the data from the smoking, alcohol consumption, physical activity, diet quality, and sleep quality

columns. The y data included the diagnosis column only. The test size of the train-test split was 20% of the dataset and two neighbors were used. The education level of participants was visualized through a bar graph. Before the data could be graphed, the value counts of the education level column had to be created and stored under education counts. Matplotlib's bar function used education counts to show the most common education level of patients.

Figure 1: The Correlation Between Age and MMSE Score

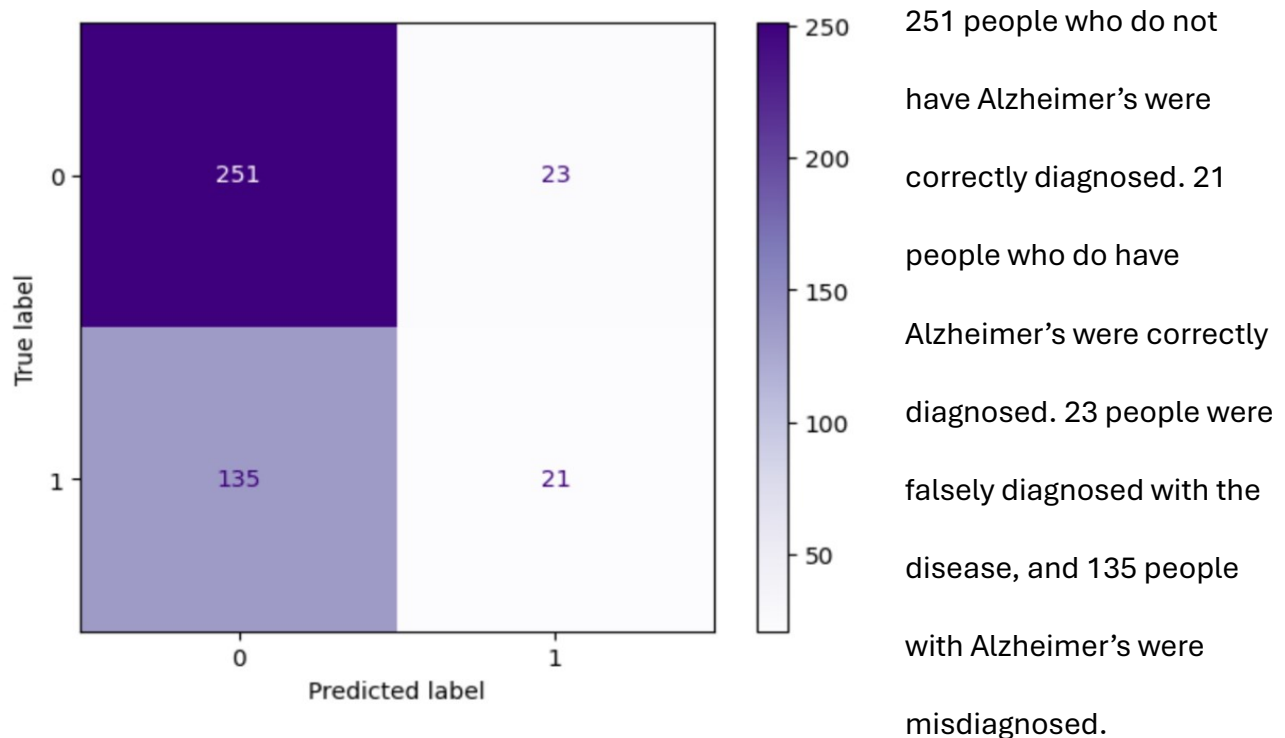


With a correlation score of -0.004, there is an extremely weak, negative, linear relationship between age and mini-mental state examination.

With MMSE's, a smaller score indicates cognitive impairment, so the correlation score reveals that as age increases, there is a minute decrease in cognitive ability. Despite the lack of correlation, there are 60-year-olds with an MMSE score of 0, as well as 90-year-olds with scores of 30. As seen above in Figure 1, by itself, age is not a predictor of MMSE scores and cognitive impairment. This finding reinforces the idea that cognitive decline is not universally tied to aging. Rather, it suggests the need to investigate other contributing factors like genetics, comorbidities, and education (Santiago and Potashkin). This finding pushes against ageist

assumptions and emphasizes the importance of personalized assessments of cognitive ability.

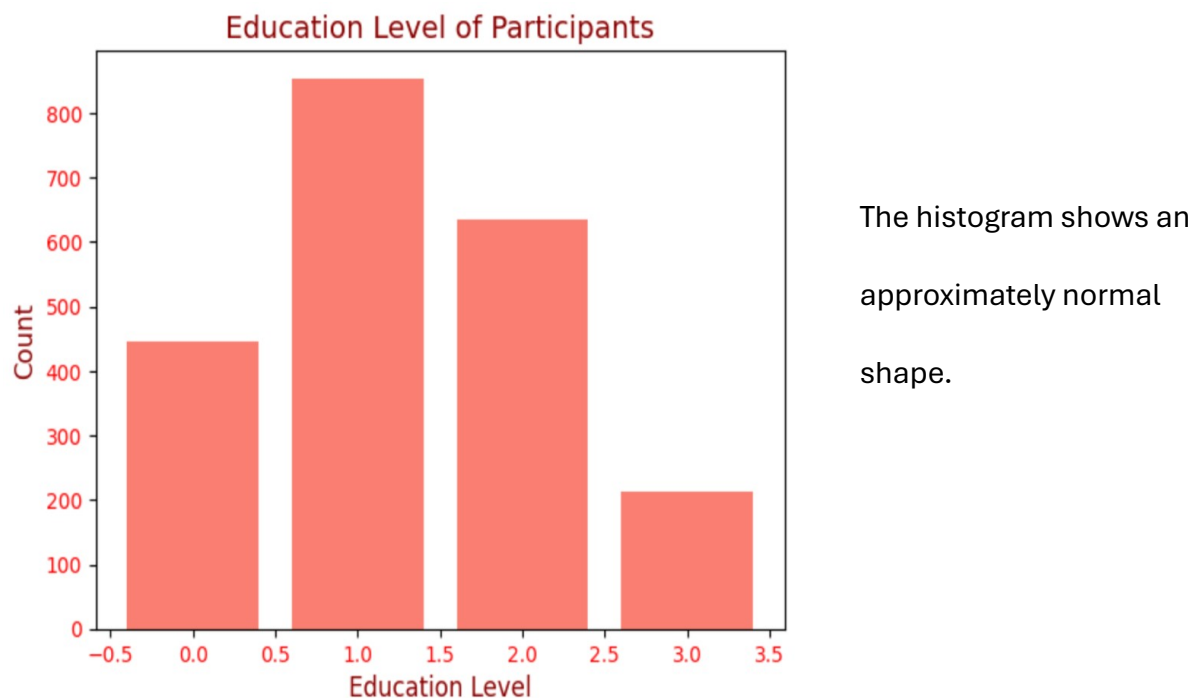
Figure 2: How Smoking, Drinking, Exercise, Diet, and Sleep Effect Diagnosis



Lifestyle factor columns within this dataset were put into a correlation matrix along with all other columns. All lifestyle factors had non-zero correlations and were therefore selected for this algorithm. Hyperparameter tuning was done, but the accuracy scores and confusion matrix changed minimally. These results reveal a high specificity, but very poor sensitivity. This means that the model can reliably identify healthy individuals based on lifestyle factors but performs poorly in detecting true Alzheimer's disease patients. Therefore, this machine learning algorithm is unreliable as a screening tool. This K Nearest

Neighbors model displays that lifestyle factors can predict who does not have Alzheimer's disease, but it cannot accurately predict patients who do have Alzheimer's Disease.

Figure 3: Education Levels of Alzheimer's Disease Dataset Participants



The dataset used a continuous variable as a categorical variable. 0 indicated no education. 1 indicated a high school diploma. 2 indicated a bachelor's degree, and 3 was a degree beyond bachelors. Most participants had a high school diploma, and a bachelor's degree was second most common. The least common degree obtained by possible Alzheimer's patients was a degree beyond that of an undergraduate. The distribution may have implications on the cognitive reserve and the cognitive reserve theory. The theory states that higher education levels can buffer against cognitive decline (Pettigrew and Soldan). The smaller number of advanced degrees obtained by Alzheimer's patients could suggest that education does provide some protection against Alzheimer's Disease.

However, since education is a lone factor in these findings, the relationship should not be overgeneralized.

The analysis of Alzheimer's disease data reveals the multifaceted nature of the condition and the various factors that may influence its onset and progression. The observed negligible correlation between age and MMSE scores reinforces the idea that aging, alone, is not a sufficient indicator of cognitive decline. This insight challenges common stereotypes and underlines the importance of considering individual variability and risk factors. The investigation into lifestyle factors through a K-Nearest Neighbors model further stresses this complexity. These results suggest that lifestyle alone cannot be used to screen for Alzheimer's, though it can guide preventive strategies and public health interventions aimed at lowering risk in healthy individuals. The distribution of education levels among participants sheds light on the potential protective role of cognitive reserve. Consistent with cognitive reserve theory, individuals with higher educational attainment may develop more robust neural networks that can delay the onset of Alzheimer's. The lower frequency of advanced degrees among Alzheimer's patients in the dataset may support this hypothesis. However, education should be viewed as a piece of the puzzle and not as a lone determinant of cognitive health.

From a public health and policy perspective, these findings suggest that early intervention strategies should focus on modifiable risk factors, such as promoting healthy lifestyle choices and improving access to quality education across the lifespan. Policies that reduce health disparities, especially among underserved communities, could play a crucial role in mitigating Alzheimer's risk and ensuring earlier, more accurate diagnoses.

These findings emphasize that understanding and addressing Alzheimer's disease requires a multidimensional approach, one that incorporates not only clinical measurements like MMSE, but also lifestyle, educational background, and possibly genetic data. This approach can inform early intervention, personalized treatment plans, and more equitable public health policies. It highlights the importance of continued research and data analysis to refine predictive models, support patients effectively, and reduce disparities in diagnosis and care.

Works Cited

- “Data, Assessment, and Analysis.” *Alzheimer’s Association*, 2015,
www.alz.org/professionals/publichealth/public-health-topics/data-assessment-analysis. Accessed
11 Apr. 2025.
- Rabie El Kharoua. “Alzheimer’s Disease Dataset.” *Kaggle.com*, 2024,
www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset?resource=download.
Accessed 11 Apr. 2025.
- Health Direct. “Mini Mental State Examination (MMSE).” *Healthdirect*, 2024,
www.healthdirect.gov.au/mini-mental-state-examination-mmse. Accessed 25 Apr. 2025.
- Pettigrew, Corinne, and Anja Soldan. “Defining Cognitive Reserve and Implications for Cognitive
Aging.” *Current Neurology and Neuroscience Reports*, vol. 19, no. 1, Jan. 2019,
<https://doi.org/10.1007/s11910-019-0917-z>. Accessed 25 Apr. 2025.
- Rabie El Kharoua. “Alzheimer’s Disease Dataset.” *Kaggle.com*, 2024,
www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset?resource=download.
Accessed 11 Apr. 2025.
- Santiago, Jose A., and Judith A. Potashkin. “The Impact of Disease Comorbidities in Alzheimer’s
Disease.” *Frontiers in Aging Neuroscience*, vol. 13, no. 1, 2021, p. 631770,
pubmed.ncbi.nlm.nih.gov/33643025/, <https://doi.org/10.3389/fnagi.2021.631770>. Accessed 2
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